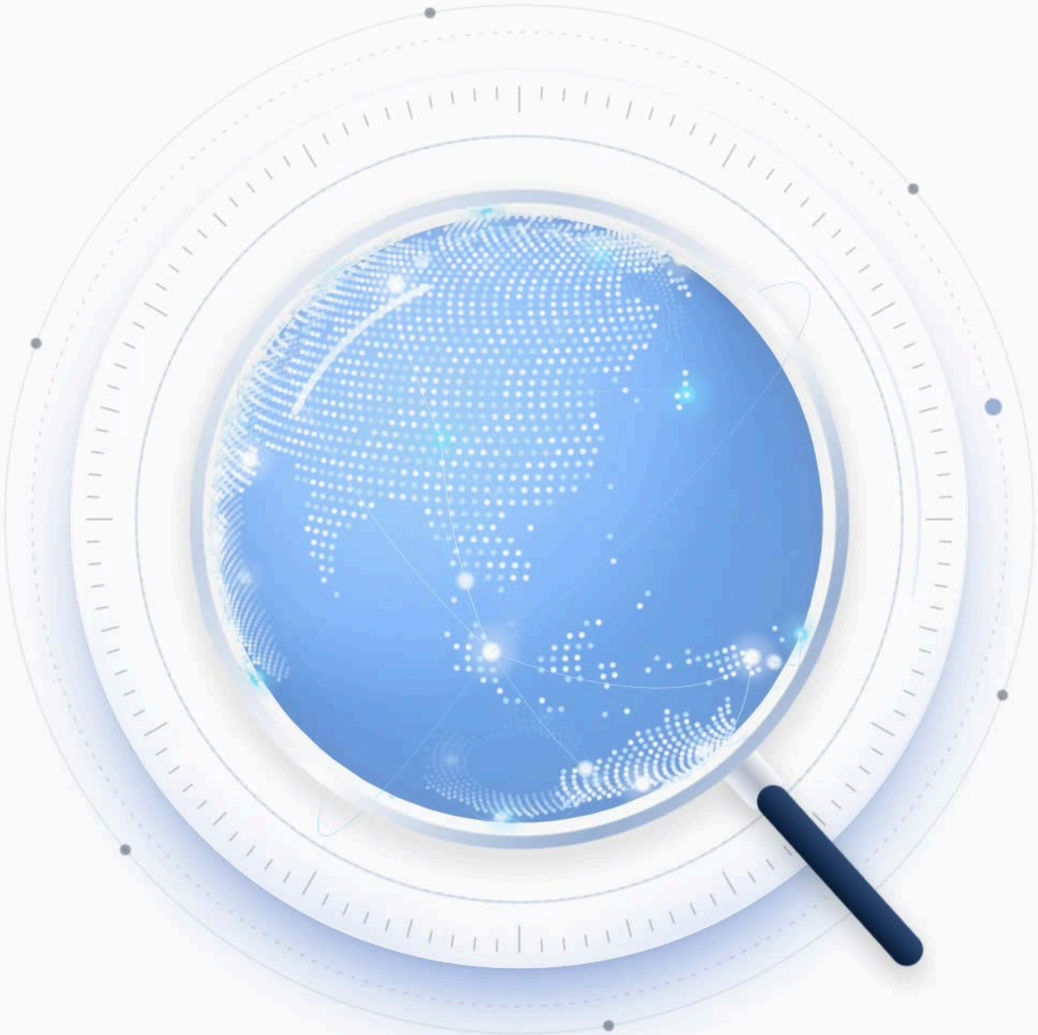
NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

# URECAMatch

## Product Showcase & Live Demo




Developer Team

**Ian Tan, Robin Teoh, Benjamin Ng, Edwin Tan Qi Lun, Nicholas Ng**


Date

**3 November 2026**

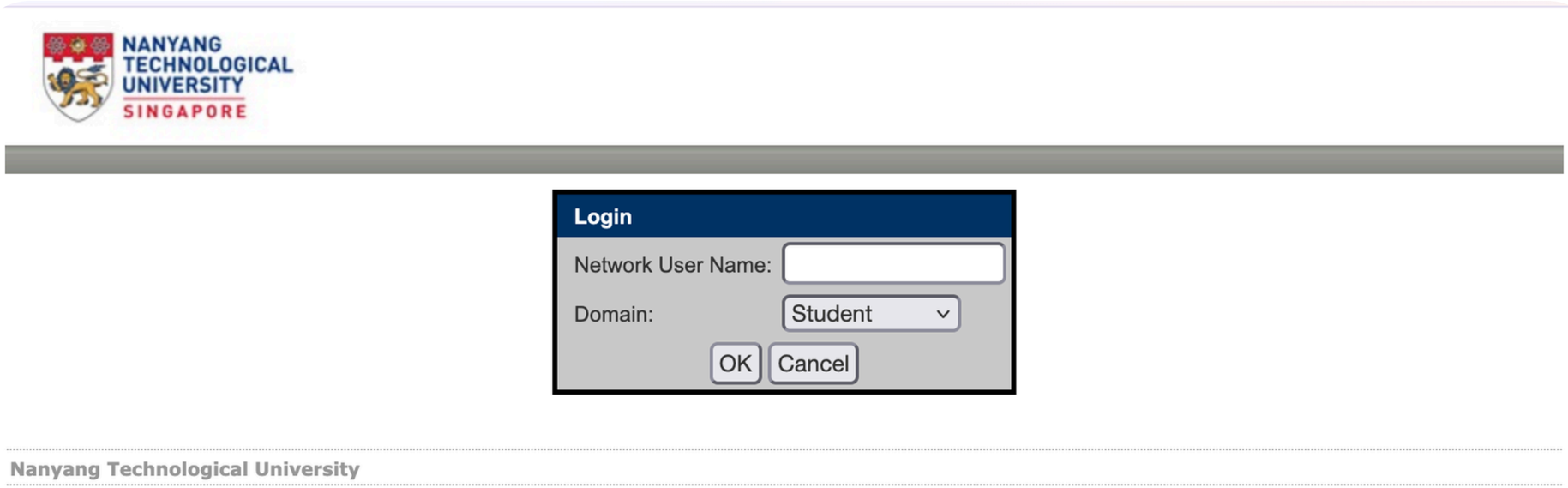
# What is URECA?

> **… acronym for Undergraduate Research Experience on CAmpus, is a university wide undergraduate research programme established in 2004 to cultivate a research culture and nurture research capabilities early in undergraduates' university education.**

- Year-long research programme undertaken by NTU Undergraduates from all faculties
- Each student is mentored by a Project Supervisor, usually a faculty staff member

# What gap are we trying to solve?

# Current URECA Portal



# Current URECA Portal

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

## List of URECA Project Proposals

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ADM ASE CCEB CEE CoB (NBS) EEE IGS MAE MSE NIE RSIS SBS SCBE SCSE SPMS SSS SoH WKW SCI All

| School | Project ID | Type | Title | Supervisor |
|---|---|---|---|---|
| ADM | ADM26026 | URECA | Personalized AI Assistant for Hand-Drawn Animation: Local ComfyUI Workflow for Style-Consistent Production | Asst Prof Ahmad Ismail Saleh |
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1. this list continues for over 1000+ entries and has no search function, by the way

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2. only way for student to apply for a project is to send an application to the professor's email

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## List of URECA Project Proposals

3. Supervisors can post multiple projects → managing all their applicants via email can be tedious

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4. Admins have zero control or information over the entire application process

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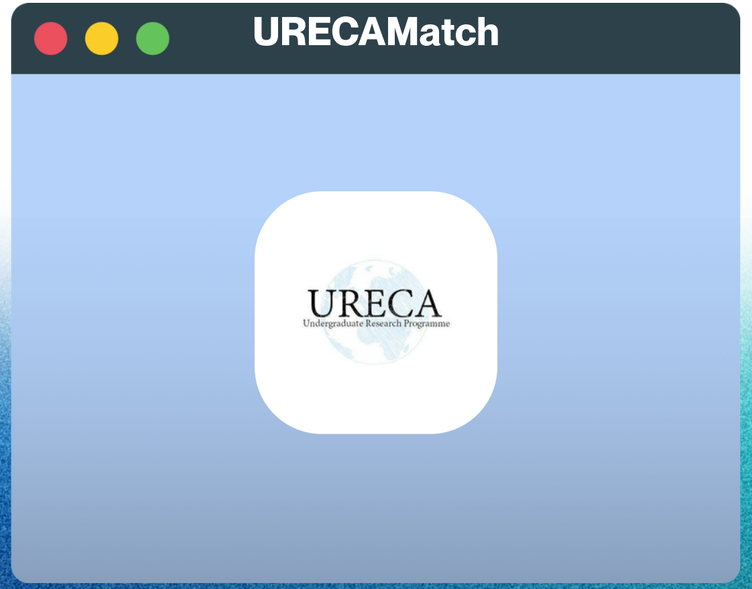
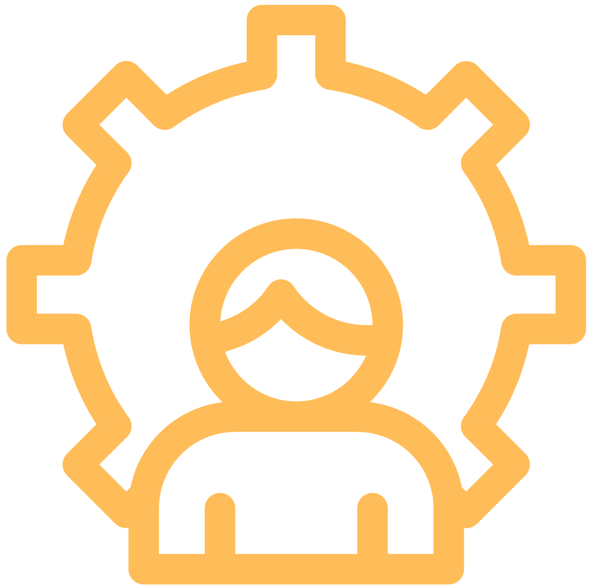
# Introducing the URECAMatch Web Portal



# Target Users



## Admin

URECA Management Staff

## Supervisor

Research Project Mentors

## Student

NTU Undergraduates

# The Problems We Solve

## and the features we offer

### 01.

**Provide a unified management tool for URECA Administrators**

Empower Admins with greater control in arbitrating project assignments, managing the application process, and streamlining workflows.

### 02.

**Centralized platform for Supervisors to manage applicant information**

Access student details, CVs, Statement of Interests, all organized by project, all in one place
No more tedious searching through email inboxes.

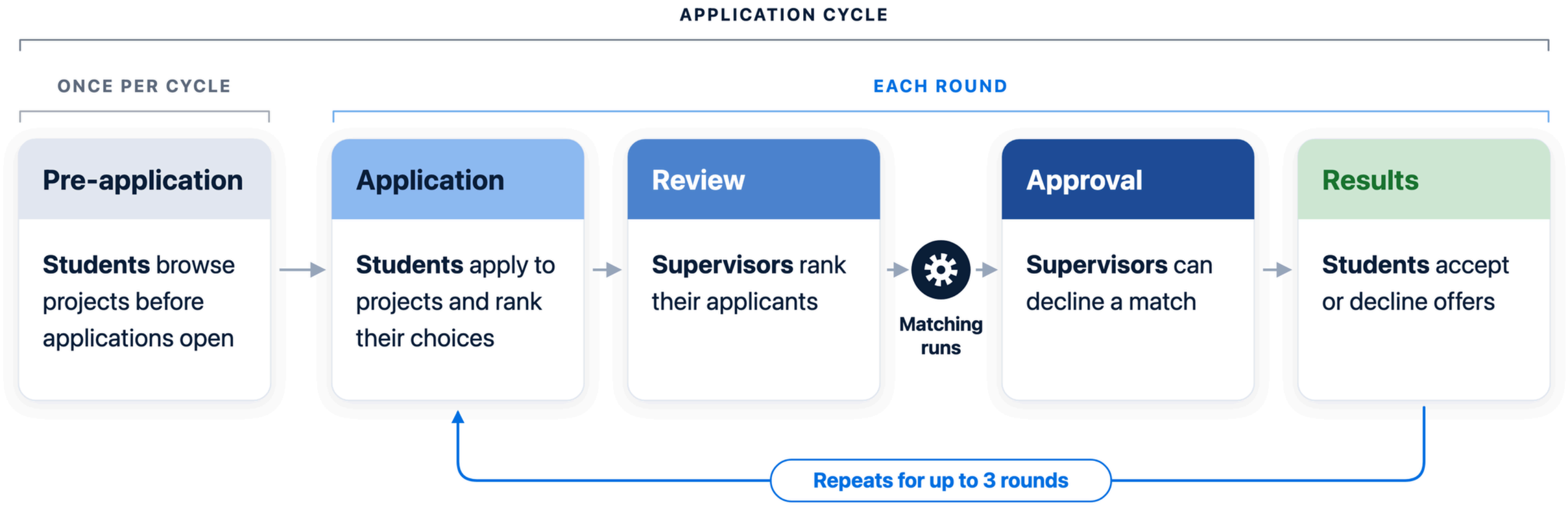
### 03.

**A more structured application process for Students**

Application cycle & rounds are well-defined, with clear deadlines for application submisstions and results announcements.

# Application Cycle

URECAMatch



# System Architecture

URECAMatch

PRESENTATION · BOUNDARY

APPLICATION · CONTROL

DATA · ENTITY

## Web client
Next.js 16, React 19

**Students**
Browse, apply, rank, respond

**Supervisors**
List projects, rank applicants

**Admins**
Set up cycles, override, audit

HTTPS

signed token

## API server
FastAPI, Python 3.12

**Guards**
Check sign-in and role

**Routers**
Enforce every rule and deadline

**Services**
Matching, notifications, audit, AI score

Service key

## Data
Supabase

**Store**
Supabase

**Postgres**
Records and audit log

**Auth and Storage**
Passwords, documents, photos

**Vercel Cron**
Daily schedule

**Resend**
E-mail delivery

**Gemini**
AI score and tips

THIRD-PARTY SERVICES/API

# URECAMatch Web Portal

# Live Demo